

Time-Series Sepsis Onset Prediction (LSTM vs. Transformer) + Topic Modeling on Notes

Goal

From temporal vitals/labs (hourly for 24–48h) predict **sepsis onset in next 6–12h**. Compare:

- **LSTM/GRU** sequence models
vs.
- **Transformer encoder** (small).

Add a parallel **NLP side task**: do **topic modeling with gensim LDA** on short nurse/physician notes to discover themes; optionally derive **note-topic features** and fuse with time-series model.

Data (synthetic; medium)

- Generate ~8k episodes with 24–48 time steps and 6–10 channels (HR, SBP, DBP, RR, Temp, SpO2, WBC, Lactate).
- Labels: sepsis_onset_within_12h based on deteriorating patterns (e.g., rising lactate, tachycardia).
- Notes: brief hourly/shift notes with phrases aligned to deterioration (e.g., “increasing lactate”, “pressors started”).

How to create

Use the provided script below (produces series.npz with (N, T, F), labels, and notes.csv with short texts).

Starter code (data synth + models)

Provided separately

Topic modeling add-on (gensim LDA)

Tokenize notes.csv, build dictionary + corpus, train LDA (k=5–10), add per-episode topic proportions as features into a late-fusion logistic regression or as additional input to the sequence model.

General Guidance to Students

- **Metrics**: favor **ROC-AUC/PR-AUC** for imbalanced outcomes, **macro-F1** for multi-class text, **mAP** for detection, and per-class breakdowns.
- **Comparisons**: make at least **two strong baselines** and **one ablation** per project.
- **Healthcare-specific models to try (optional)**:
 - **Embeddings/NLP**: *Bio_ClinicalBERT*, *BioBERT*, *SapBERT*, *PubMedBERT*
 - **Summarization/Q&A**: *pegasus-pubmed* (summ), *BioGPT* (exploration)
 - **spaCy**: try **scispaCy** models for biomedical entities

Evaluation rubric (100 pts)

- Correct time-series framing & label horizon: **20**
- LSTM vs Transformer comparison with robust metrics (ROC-AUC/PR-AUC): **30**
- Topic modeling integration & discussion of utility: **20**
- Error/ablation analysis (window length, features): **20**
- Reproducibility/report: **10**